Reproducing and Evaluating MIPIC: Matryoshka Representation Learning with SIA and PIC

CS 613: Natural Language Processing – Assignment 1 Report

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Abstract

Fixed-dimensional text representations can require more memory and computation than necessary when downstream applications have different resource constraints. Matryoshka Representation Learning (MRL) addresses this problem by learning nested representations inside a single embedding, so that smaller prefixes can still be useful for downstream tasks. MIPIC extends MRL by organizing information across representation dimensions and Transformer layers using Self-Distilled Intra-Relational Alignment (SIA) and Progressive Information Chaining (PIC). This report is a reproduction study of MIPIC. We reimplement the method on a TinyBERT-6L backbone from the paper's description together with the authors' reference implementation, train MRL, MRL + SIA, MRL + PIC and MIPIC under one controlled protocol, and evaluate each with a linear probe at seven nested dimensions. Two departures from the original are unavoidable and are documented throughout: the downstream evaluation is a single three-class tweet sentiment task rather than the seven-task TweetEval average, and the evaluation files shipped with the reference implementation are the Kaggle Tweet Sentiment Extraction dataset rather than the TweetEval benchmark. We therefore test whether the paper's qualitative claims survive a controlled reimplementation, using macro-F1 to measure how much performance is retained as the representation is compressed.

1. Introduction

1.1 Motivation

Modern NLP systems commonly use dense text representations with hundreds of dimensions. While these representations can capture useful semantic information, using the full representation is not always necessary for a downstream task. Smaller representations can reduce memory usage, storage requirements, and computation, which is useful when models need to operate under limited resources.

Matryoshka Representation Learning (MRL) addresses this problem by learning representations where smaller prefixes of the full embedding remain useful. This allows the same model to provide representations of different sizes without training a separate model for each size. However, simply training multiple prefixes does not guarantee that the most useful information will be placed in the earlier dimensions.

MIPIC builds on MRL by explicitly controlling how information is organized across representation dimensions and Transformer layers. It uses Self-Distilled Intra-Relational Alignment (SIA) to preserve relationships between representations of different dimensions, and Progressive Information Chaining (PIC) to pass useful information from deeper Transformer layers towards earlier layers.

In this project, we study whether these two components improve the quality of low-dimensional representations compared with standard MRL. The main goal is to determine whether useful task information can be retained even when the representation is strongly com-

pressed.

1.2 Relation with NLP

The problem is directly related to NLP because the representations being compressed contain semantic and contextual information from natural language. Given a tweet, the model must encode information that can later be used for classification.

The challenge is to retain this information even when only a small part of the full representation is used. MIPIC addresses this during representation learning instead of applying dimensionality reduction after training. SIA focuses on maintaining relationships across representation dimensions, while PIC guides the transfer of useful information across Transformer layers.

We therefore study whether these learned compact representations retain the linguistic information required for downstream text classification.

1.3 Problem Type

This project is a **reproduction study**. The underlying problem is **representation learning** evaluated through **downstream text classification**, but the deliverable is not a new method. It is a controlled reimplementation of MIPIC and a verdict on which of the paper's claims survive it.

We reimplement the method from the paper's description together with the authors' reference implementation, and we test four claims:

1. **C1.** MIPIC outperforms MRL at every nested dimension.
2. **C2.** The margin over MRL is largest at the smallest dimensions, that is, MIPIC degrades less under compression.
3. **C3.** SIA and PIC each contribute individually, so both single-component models sit between MRL and MIPIC.
4. **C4.** The reported hyperparameters (Table 7 of the paper) are sufficient to obtain this behaviour.

2. Related Work

Our work is mainly based on Matryoshka Representation Learning (MRL) (Kusupati et al., 2022) and MIPIC: Matryoshka Representation Learning via Self-Distilled Intra-Relational and Progressive Information Chaining (Huy et al., 2026).

2.1 Matryoshka Representation Learning

Matryoshka Representation Learning (MRL) (Kusupati et al., 2022) introduces a way to learn multiple useful representations inside a single embedding. Instead of training separate models for different representation sizes, MRL jointly trains representations at several nested dimensions.

For example, if the full representation has 768 dimensions, the first 128 dimensions can be used as a smaller representation while the full representation is still available. The main idea is that each smaller prefix should retain useful information for the downstream task.

The original paper evaluates MRL in several settings, including computer vision, vision-language learning, and NLP. For NLP, the authors apply MRL to BERT and show that smaller representations can remain useful compared with separately trained fixed-dimensional representations.

Such representations could also be useful in future NLP systems deployed on resource-constrained edge devices. If the approach is ineffective, smaller representations may lose useful information, reducing the benefit of compression and requiring larger representations for downstream tasks.

The MRL baseline and the MIPIC reference implementation are available, which provides a basis for comparing the proposed method with its baseline under a controlled experimental setting.

2.2 MIPIC

MIPIC (Huy et al., 2026) extends MRL by focusing on how information is organized both across representation dimensions and across Transformer layers.

It introduces two main components. **Self-Distilled Intra-Relational Alignment (SIA)** aligns relational information between the full representation and its truncated versions. This is intended to help smaller representations retain the structure present in the full representation.

Progressive Information Chaining (PIC) transfers useful information from deeper Transformer layers towards earlier layers. This provides additional guidance

Dim.	SimCSE	MRL	ESE	MIPIC
16	35.12	53.66	52.71	60.57
32	46.06	59.08	59.17	63.32
64	49.72	60.22	65.16	65.36
128	55.89	66.71	66.75	67.61
256	63.54	68.03	69.93	69.29
512	66.47	68.25	69.32	70.43
768	70.16	69.10	70.02	70.89

Table 1: Reported TweetEval F1 scores for TinyBERT-6L from the MIPIC paper at different representation dimensions.

to intermediate representations.

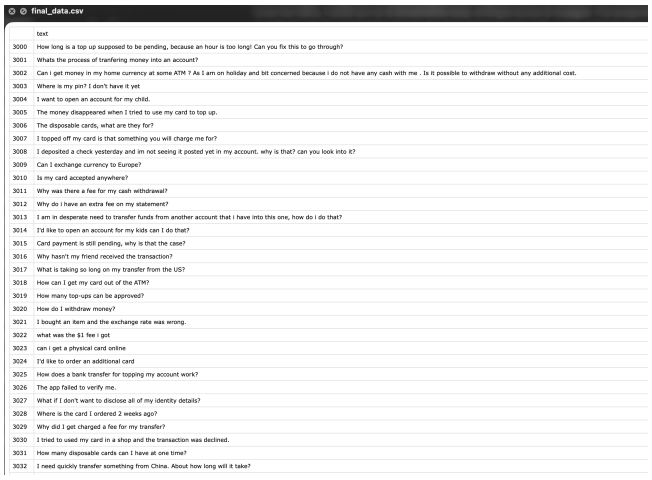
Together, SIA and PIC are designed to improve the preservation of useful information under dimensional compression. The MIPIC paper evaluates this approach on several NLP tasks, including TweetEval, and directly compares it with MRL.

Our contribution is reproduction rather than extension.

Two things distinguish our runs from the published numbers in Table 1. First, all four models are trained by us under one protocol: the same corpus, schedule, batch size, gradient clipping, seed and sentence read-out, so any difference between them is attributable to the loss terms alone. The published table compares methods trained by different authors under different conditions, which is reasonable as a survey but weaker as evidence that SIA and PIC are what produce the gain. Second, we train the two single-component models, MRL + SIA and MRL + PIC, so the contribution of each term can be separated rather than inferred from the combined model. Where the paper reports its own component ablation, these two arms reproduce it under the controlled protocol above.

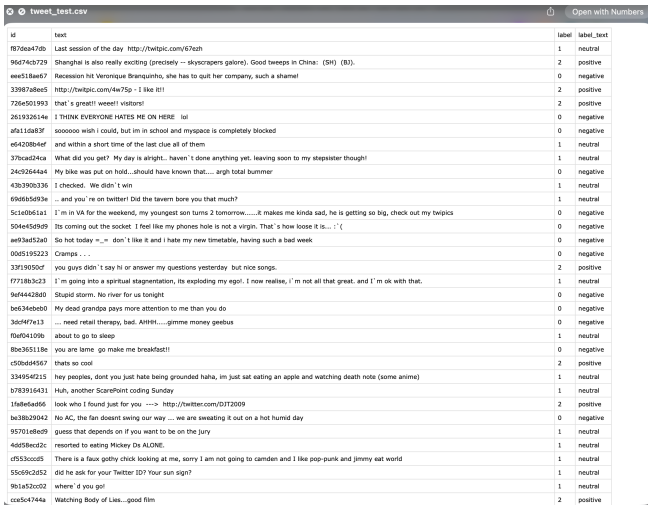
3. Datasets

Two datasets are involved, and they play different roles. The first is an unlabelled sentence pool used for contrastive training of the encoder. The second is a labelled tweet sentiment set used only at evaluation time to fit and score a linear probe. No labels are used during encoder training.


final_data.csv

	text
3000	How long is a top up supposed to be pending, because an hour is too long! Can you fix this to go through?
3001	Whats the process of tranfering money into an account?
3002	Can I get money in my home currency at some ATM ? As I am on holiday and bit concerned because I do not have any cash with me . Is it possible to withdraw without any additional cost.
3003	Where is my pin? I don't have it yet
3004	I want to open an account for my child.
3005	The money disappeared when I tried to use my card to top up.
3006	The disposable cards, what are they for?
3007	I topped off my card is that something you will charge me for?
3008	I deposited a check yesterday and im not seeing it posted yet in my account. why is that? can you look into it?
3009	Can I exchange currency to Europe?
3010	Is my card accepted anywhere?
3011	Why was there a fee for my cash withdrawal?
3012	Why do i have an extra fee on my statement?
3013	I am in desperate need to transfer funds from another account that i have into this one, how do i do that?
3014	I'd like to open an account for my kids can I do that?
3015	Card payment is still pending, why is that the case?
3016	Why hasn't my friend received the transaction?
3017	What is taking so long on my transfer from the US?
3018	How can I get my card out of the ATM?
3019	How many top-ups can be approved?
3020	How do I withdraw money?
3021	I bought an item and the exchange rate was wrong.
3022	what was the $1 fee I got
3023	can i get a physical card online
3024	I'd like to order an additional card
3025	How does a bank transfer for topping my account work?
3026	The app failed to verify me.
3027	What if I don't want to disclose all of my identity details?
3028	Where is the card I ordered 2 weeks ago?
3029	Why did I get charged a fee for my transfer?
3030	I tried to used my card in a shop and the transaction was declined.
3031	How many disposable cards can I have at one time?
3032	I need quickly transfer something from China. About how long will it take?

Figure 1: Preparation and verification of the training corpus.


tweet_test.csv — Open with Numbers

id	text	label	label_text
f87dea47db	Last session of the day http://twitpic.com/67ezh	1	neutral
96d74cb729	Shanghai is also really exciting (precisely -- skyscrapers galore). Good tweeps in China: (SH) (BJ).	2	positive
eee518ae67	Recession hit Veronique Branquinho, she has to quit her company, such a shame!	0	negative
33987a8ee5	http://twitpic.com/4w75p - I like it!!	2	positive
726e501993	that's great!! weee!! visitors!	2	positive
261932614e	I THINK EVERYONE HATES ME ON HERE lol	0	negative
afa11da83f	soooooo wish i could, but im in school and myspace is completely blocked	0	negative
e64208b4ef	and within a short time of the last clue all of them	1	neutral
37bcad24ca	What did you get? My day is alright.. haven't done anything yet. leaving soon to my stepsister though!	1	neutral
24c92644a4	My bike was put on hold...should have known that.... argh total bummer	0	negative
43b390b336	I checked. We didn't win	1	neutral
69d6b5d93e	.. and you're on twitter! Did the tavern bore you that much?	1	neutral
5c1e0b61a1	I'm in VA for the weekend, my youngest son turns 2 tomorrow.....it makes me kinda sad, he is getting so big, check out my twipics	0	negative
504e45d9d9	Its coming out the socket I feel like my phones hole is not a virgin. That's how loose it is... :'(	0	negative
ae93ad52a0	So hot today =_= don't like it and i hate my new timetable, having such a bad week	0	negative
00d5195223	Cramps . . .	0	negative
33f19050cf	you guys didn't say hi or answer my questions yesterday but nice songs.	2	positive
f77188c3c23	I'm going into a spiritual stagnentation, its exploding my ego!. I now realise, i'm not all that great. and I'm ok with that.	1	neutral
9af4428d0	Stupid storm. No river for us tonight	0	negative
be634ebeb0	My dead grandpa pays more attention to me than you do	0	negative
3dcf4f7e13	... need retail therapy, bad. AHHH.....gimme money geebus	0	negative
f0af04109b	about to go to sleep	1	neutral
8be36511Be	you are lame go make me breakfast!!	0	negative
c50bdd4567	thats so cool	2	positive
334954f215	hey peoples, dont you just hate being grounded haha, im just sat eating an apple and watching death note (some anime)	1	neutral
b783916431	Huh, another ScarePoint coding Sunday	1	neutral
1fa8e6ad66	look who I found just for you ---> http://twitter.com/DJT2009	2	positive
be38b29042	No AC, the fan doesnt swing our way ... we are sweating it out on a hot humid day	0	negative
95701e8ed9	guess that depends on if you want to be on the jury	1	neutral
4dd58ecd2c	resorted to eating Mickey Ds ALONE.	1	neutral
cf553cccd5	There is a faux gothy chick looking at me, sorry I am not going to camden and I like pop-punk and jimmy eat world	1	neutral
55c69c2d52	did he ask for your Twitter ID? Your sun sign?	1	neutral
9b1a52cc02	where' d you go!	1	neutral
cce5c4744a	Watching Body of Lies...good film	2	positive

Figure 2: Tweet sentiment test-set evaluation.

3.1 Availability and Size

Training corpus. Encoder training uses the pooled sentence corpus supplied with the MIPIC reference implementation (`data/final_data.csv`), containing 20,240 sentences drawn from seven sources. The composition is given in Table 2. This matches the pooling described in Appendix A.2 of the MIPIC paper. Sentences are short: the mean length is 12.9 whitespace tokens, the median is 11, and the longest is 79.

Evaluation data. Downstream evaluation uses the labelled tweet files shipped with the reference implementation (`tweet_train.csv`, `tweet_test.csv`): a three-class sentiment task (`negative`, `neutral`, `positive`) with 26,732 training tweets and 3,432 test tweets. The class distribution is mildly imbalanced in both splits: the

Source	Sentences
BANKING77	3,000
TWEET_EVAL_SENT	3,000
MRPC	3,000
DEEHAN1866/WIC	3,000
DGEM_FORMAT	3,000
STSB	2,893
SICK	2,347
Total	**20,240**

Table 2: Composition of the contrastive training corpus.

training split is 40.0% neutral, 31.3% positive and 28.7% negative, and the test split is 40.1% neutral, 31.3% positive and 28.6% negative.

These files are *not* the TweetEval benchmark. Their row identifiers and text match the Kaggle Tweet Sentiment Extraction dataset exactly, including its first training row, and their sizes are consistent with that dataset (27,481 / 3,534) after empty rows are dropped. The public TweetEval sentiment split is 45,615 / 12,284 and carries no such identifiers. This matters for a reproduction: the column labelled TweetEval in Table 1 and the column we produce are measured on different data, so they are not comparable as absolute numbers.

The evaluation is also a single task rather than the seven-task TweetEval average. This is a deliberate scope reduction, since each additional task requires a separate probe over all seven nested dimensions for all four models. Taken together, the two points mean our reproduction tests whether the paper's *ordering and trend* survive, not whether its F1 values are recovered. Comparisons *between* our four models remain valid, because all four use the identical protocol and the identical data.

Neither dataset is newly collected by this project. Both are used as supplied with the reference implementation.

3.2 Data Collection and Labeling

Since the data are supplied with the reference implementation, no new data collection or labeling is needed. The data preparation mainly consists of obtaining the files, preprocessing the tweets, and preparing the training and evaluation inputs. The evaluation data are already labeled, so no additional manual or LLM-based annotation is required.

```
Training completed in 964.3s (16.07 min)
Checkpoint saved to /kaggle/working/mipic_results_all/checkpoints/mrl/model.pt
  Epoch 1/5: mrl_loss: 0.2008, task_loss: 0.0126
  Epoch 2/5: mrl_loss: 0.0048, task_loss: 0.0002
  Epoch 3/5: mrl_loss: 0.0028, task_loss: 0.0001
  Epoch 4/5: mrl_loss: 0.0034, task_loss: 0.0001
  Epoch 5/5: mrl_loss: 0.0020, task_loss: 0.0001

Running TweetEval Linear Probe evaluation for mrl...
Held-out probe preferred epoch 4/5; evaluating that checkpoint too.
Evaluation completed in 106.4s (1.77 min)

================================================================================
====
TWEETEVAL EVALUATION RESULTS ACROSS ALL 7 MRL DIMENSIONS
================================================================================
====
Dimension  |   MRL (Acc / F1)
--------------------------------
    16     |  56.64% /  56.46%
    32     |  59.99% /  59.99%
    64     |  62.67% /  62.82%
   128     |  65.36% /  65.57%
   256     |  66.61% /  66.82%
   512     |  67.40% /  67.61%
   768     |  67.02% /  67.24%
================================================================================
```

Figure 3: MRL baseline training and evaluation.

4. Experimental Plan

4.1 Experimental Setting

We use TinyBERT-6L as the backbone. Representations are evaluated at seven nested dimensions:

$$\{16, 32, 64, 128, 256, 512, 768\}.$$

We compare four configurations: MRL, MRL + SIA, MRL + PIC, and the complete MIPIC model. All four configurations use the same training corpus and evaluation protocol. The only difference between the runs is the set of loss components enabled.

4.2 Train / Development / Test Protocol

The encoder is trained on the supplied unlabeled training corpus. For downstream evaluation, a linear probe is trained using the supplied 26,732-example training split and evaluated on the 3,432-example test split. No separate development split is used.

The same probe procedure is applied independently at each of the seven representation dimensions for all four model configurations.

4.3 Hyperparameter Selection

We do not perform grid search or automated hyperparameter optimization. Instead, we used the combination of values specified in the MIPIC paper and its provided implementation and keep them fixed across the four configurations.

Parameter	Value
Backbone	TinyBERT-6L
MRL dimensions	16, 32, 64, 128, 256, 512, 768
Batch size	16
Learning rate	2e-5
Optimizer	AdamW (cosine + 10% warmup)
Epochs	5
Temperature	0.05
α	0.4
Random seed	42

```
attn_loss = torch.stack(attn_terms).mean()
cka_loss = torch.stack(cka_terms).mean()
sia_loss = self.w_attn * attn_loss + self.w_cka * cka_loss
```

(a) SIA loss

```
# Normalize for cosine similarity
u_proj = F.normalize(u_proj, dim=-1)
v_tgt = F.normalize(v_tgt, dim=-1)

# Under DDP, keep local queries but gather all target keys so the
# InfoNCE denominator uses the global effective batch.
gathered_targets = torch.cat(gather_with_grad(v_tgt), dim=0)
logits = torch.matmul(u_proj, gathered_targets.T) / temperature

# Positive for local query i is the corresponding global target i.
labels = global_batch_labels(u_proj.size(0), u_proj.device)

# InfoNCE loss (cross entropy over in-batch negatives)
loss = F.cross_entropy(logits, labels)

return loss
```

(b) PIC loss

Figure 4: Training losses for the SIA and PIC components.

4.4 Evaluation Metric

We report both accuracy and macro-F1. Macro-F1 is the primary metric because it gives equal weight to the three sentiment classes. Results are reported independently at each of the seven representation dimensions.

4.5 System / Demo Scope

The project focuses on implementing and evaluating the MIPIC training framework rather than building a separate application. The final demonstration shows the training and evaluation pipeline and the use of the same encoder at different representation dimensions.

5. Project Management

5.1 Timeline

The project is divided into stages covering baseline implementation, component development, integration, experiments, and report writing.

5.2 Computational Resources

The experiments require 2x T4-GPU from Kaggle or a local MPS or CUDA based GPU for training the TinyBERT-6L model and running the different model configurations. They use up almost all of the GPU capacity on both local training and on Kaggle.

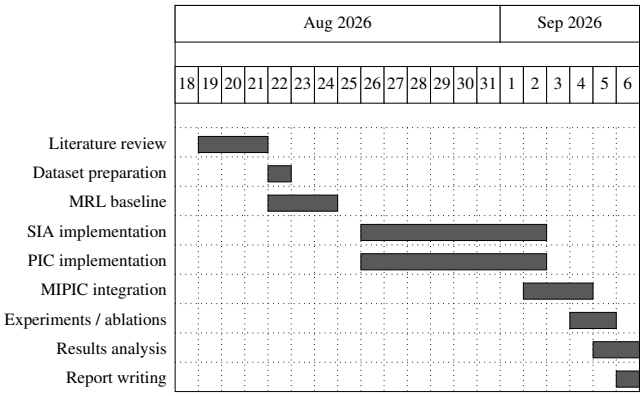


Figure 5: Assignment timeline from August 18 to September 6, 2026.

5.3 Definition of Success

We consider the project successful if we can:

- reproduce the MRL and MIPIC training pipelines;
- obtain valid representations at all selected dimensions;
- complete the planned ablation experiments;
- measure the contribution of SIA and PIC; and
- compare our results with the main trend reported by MIPIC.

5.4 Biggest Risk

The main risk is that the full MIPIC training setup may require more GPU memory or training time than is available for the project. This can limit the number of experiments and hyperparameter settings that can be tested. To reduce this risk, we first validate the complete pipeline on a small subset of the data.

5.5 Task Distribution

The project is divided into four teams, with each team responsible for a specific part of the MIPIC pipeline.

- **Team 1 – Saharsh and Shivamani: MRL Backbone.** Build and validate the MRL backbone and

establish the baseline training pipeline. After validation, scale the baseline training and perform the required tuning.

- **Team 2 – Eshwar and Dhruthika: MRL + SIA.** Implement the Self-Distilled Intra-Relational Alignment (SIA) component and integrate it with the MRL backbone.
- **Team 3 – Aniruddh and Akash: MRL + PIC.** Implement the Progressive Information Chaining (PIC) component and integrate it with the MRL backbone.
- **Team 4 – Anvesh and Manoj: MIPIC Integration and Documentation.** Develop the MIPIC wrapper that combines MRL, SIA, and PIC. This team also handles the final comparative experiments, and the project report.

6. Experiments and Results

6.1 Reproduction

We evaluated the four trained models on the supplied three-class tweet sentiment data. The evaluation uses the full 26,732-row training split to fit a Logistic Regression probe and the full 3,432-row test split for scoring. The seven nested dimensions are evaluated from the same frozen encoder outputs.

The resulting values are not directly comparable with the paper's reported TweetEval aggregate because our evaluation uses one sentiment dataset rather than the paper's multi-task evaluation, and the supplied files are different from the TweetEval benchmark. We therefore use the paper only as a qualitative reference and compare the four models under the same local protocol.

6.2 Ablation Study

We compare the MRL baseline with the two single-component models, MRL + SIA and MRL + PIC, and with the complete MIPIC model. Because all four models use the same corpus, schedule, probe data and evaluation procedure, the comparison isolates the observed effect of adding each component under this setup.

The MIPIC model has the highest macro-F1 at 16 and 32 dimensions, with gains of 2.14 and 0.52 points over MRL, respectively. From 64 dimensions onward, MRL has the highest macro-F1 among the four models. MRL + SIA and MRL + PIC remain below MRL at every evaluated dimension.

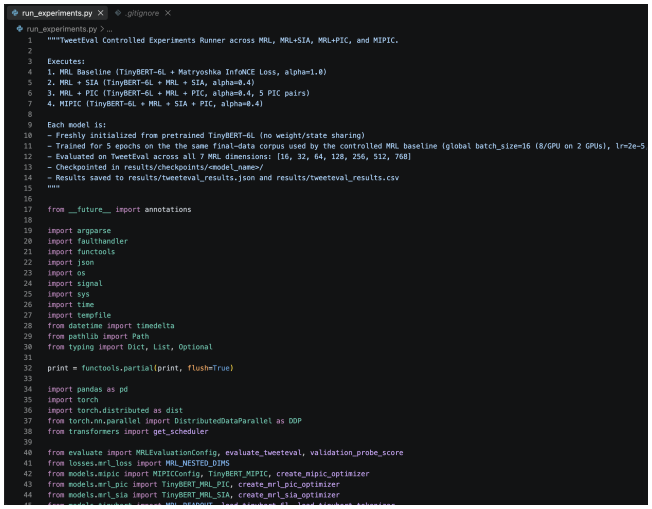


Figure 6: Complete experimental pipeline

```
Training completed in 2644.7s (44.08 min)
Checkpoint saved to /kaggle/working/mipic_results_all/checkpoints/mipic/model.pt
  Epoch 1/5: total_loss: 1.0857, mipic_loss: 1.0857, mrl_loss: 0.2314, sia_loss: 0.1634, pic_los
s: 1.4918
  Epoch 2/5: total_loss: 0.2758, mipic_loss: 0.2758, mrl_loss: 0.0225, sia_loss: 0.0703, pic_los
s: 0.3744
  Epoch 3/5: total_loss: 0.1685, mipic_loss: 0.1685, mrl_loss: 0.0136, sia_loss: 0.0550, pic_los
s: 0.2168
  Epoch 4/5: total_loss: 0.1361, mipic_loss: 0.1361, mrl_loss: 0.0101, sia_loss: 0.0506, pic_los
s: 0.1696
  Epoch 5/5: total_loss: 0.1279, mipic_loss: 0.1279, mrl_loss: 0.0074, sia_loss: 0.0493, pic_los
s: 0.1589
```

Figure 7: MIPIC training and evaluation pipeline.

6.3 Findings

The results show two different patterns. In absolute macro-F1, the MIPIC model is strongest at the two smallest dimensions, reaching 58.60 at 16 dimensions and 60.51 at 32 dimensions. At 64 dimensions and above, the MRL baseline is consistently stronger, with 67.24 at 768 dimensions compared with 64.64 for MIPIC.

The compression trend is somewhat different from the absolute ranking. From 768 to 16 dimensions, MRL drops by 10.78 F1 points, whereas MIPIC drops by 6.04 points. Thus, although MIPIC does not achieve the highest absolute score at larger dimensions in our experiment, its performance degrades less sharply under stronger compression.

The single-component models do not improve absolute macro-F1 over MRL in this setup: MRL + SIA is lower than MRL at all seven dimensions, and MRL + PIC is also lower than MRL at all seven dimensions. Consequently, the results do not support C1 or C3 as stated in Section 1. C2 is partially supported by the smaller performance drop of MIPIC under compression. C4 cannot be assessed independently from the

Dim.	MRL		MRL + SIA		MRL + PIC		MIPIC	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
16	56.64	56.46	53.58	53.26	54.81	54.27	58.71	58.60
32	59.99	59.99	55.71	55.16	59.99	59.85	60.58	60.51
64	62.67	62.82	56.73	56.16	61.54	61.62	61.95	61.83
128	65.36	65.57	57.78	57.30	62.41	62.55	62.67	62.65
256	66.61	66.82	58.80	58.37	63.00	63.21	62.91	62.92
512	67.40	67.61	59.35	58.99	64.02	64.27	63.90	64.01
768	67.02	67.24	59.85	59.53	64.54	64.82	64.54	64.64

Table 3: Final tweet-sentiment probe results. Accuracy and macro-F1 are reported as percentages.

evaluation differences described in Section 3.

6.4 Comparison with the Paper

The paper reports higher absolute MIPIC scores on its multi-task TweetEval benchmark. Our absolute values are lower, and the two sets of numbers should not be compared directly because the evaluation task and dataset differ.

The more relevant comparison is the qualitative behaviour across dimensions. Our MIPIC model performs best at the two smallest dimensions, while MRL performs better at larger dimensions. Thus, the experiment provides partial evidence for the paper's claim that MIPIC can improve performance under stronger compression, but it does not reproduce the paper's claim that MIPIC is consistently superior to MRL at all dimensions.

7. Conclusion

This project reimplements MIPIC on TinyBERT-6L and evaluates MRL, MRL + SIA, MRL + PIC and MIPIC under a common protocol. On the supplied three-class tweet sentiment evaluation, MIPIC improves over MRL at 16 and 32 dimensions, but MRL remains stronger from 64 dimensions onward. The MIPIC model nevertheless shows a smaller performance drop between the full and most compressed representations.

The component ablations do not show an individual improvement from SIA or PIC under this evaluation. Since the evaluation dataset and task differ from the paper's reported multi-task TweetEval evaluation, these results are interpreted as evidence about the relative behaviour of the implementations rather than as a direct reproduction of the paper's absolute scores.

Member	Contribution	%
Manoj	MIPIC wrapper and Report	12.5
Anvesh	MIPIC wrapper and Report	12.5
Saharsh	MRL Baseline code	12.5
Shivamani	MRL Baseline code	12.5
Akash	PIC implementation	12.5
Aniruddh	PIC implementation	12.5
Dhruthika	SIA implementation	12.5
Eshwar	SIA implementation	12.5

Table 4: **Contribution Breakdown** across members

Code and Data

All code and experiment configurations are available at GitHub, while the main execution at Kaggle notebook and the evaluation result files are available in Kaggle at Files Dataset.

References

A. Kusupati, G. Bhatt, A. Rege, M. Wallingford, A. Sinha, V. Ramanujan, W. Howard-Snyder, K. Chen, R. van den Berg, S. Kakade, P. Jain, and A. Farhadi. Matryoshka Representation Learning. *Advances in Neural Information Processing Systems*, 2022.

P. G. Huy, H. A. Vu, M.-P. Truong, T. D. Tran, L. N. Van, T. N. Hong, and T. Le. MIPIC: Matryoshka Representation Learning via Self-Distilled Intra-Relational and Progressive Information Chaining. *Findings of the Association for Computational Linguistics: ACL 2026*, 13824–13838, 2026.

What the paper specifies (or skips)	What the provided implementation does	What our implementation does
1. Training objective. The paper defines $L = \alpha L_{\mathrm{MRL}} + (1-\alpha)(L_{\mathrm{SIA}} + L_{\mathrm{PIC}}),$ $\alpha = 0.4.$	The ordinary MRL/task loss is disabled (`loss_task = 0`). The final loss instead combines the alignment term with a separately computed InfoNCE term.	Uses the stated decomposition: $0.4L_{\mathrm{MRL}} + 0.6(L_{\mathrm{SIA}} + L_{\mathrm{PIC}}),$ with the MRL term active.
2. SIA attention alignment. SIA includes a KL-divergence loss aligning lower-dimensional attention with the full-dimensional teacher attention.	The KL loss is computed but subsequently set to zero: `kl_loss = 0`. Thus the attention-alignment branch contributes no gradient.	The KL attention loss is active and contributes to SIA.
3. SIA token selection. Top-k contextual tokens are selected using the attention of the full-dimensional teacher representation.	Top-k tokens are selected using the lower-dimensional branch's attention scores.	Selects top-k tokens from the full-dimensional teacher attention, while excluding special tokens from the contextual-token pool.
4. SIA layer coverage. For TinyBERT, the paper applies SIA across all six transformer layers: $\{1, 2, 3, 4, 5, 6\}.$	The provided BERT implementation applies SIA only at layers $\{8, 10\}.$	Applies SIA across all six TinyBERT layers: $\{1, 2, 3, 4, 5, 6\}.$
5. PIC checkpoints. For TinyBERT, the paper specifies $(1, 16) \to (2, 32) \to (3, 64) \to (4, 256)$ $\to (5, 512) \to (6, 768).$ There is no 128-dimensional PIC stage.	The provided implementation uses a shorter BERT chain: $(3, 16) \to (7, 128) \to (11, 768),$ giving only two transitions.	Uses the paper's TinyBERT checkpoint chain with five adjacent PIC transitions and no 128-dimensional stage.
6. Classification probe size. The paper's reported classification protocol uses a 3,000-example sampled TweetEval training probe.	Uses the full TweetEval training split: 26,732 examples.	Also uses the full 26,732-example TweetEval training split.

Table 5: Six major discrepancies between the MIPIC paper, the provided implementation, and our implementation. The third column reflects the current repository configuration. These are the main reasons our implementation was not able to match the values mentioned in the paper. But we tried to keep our implementation as close as possible to the paper.